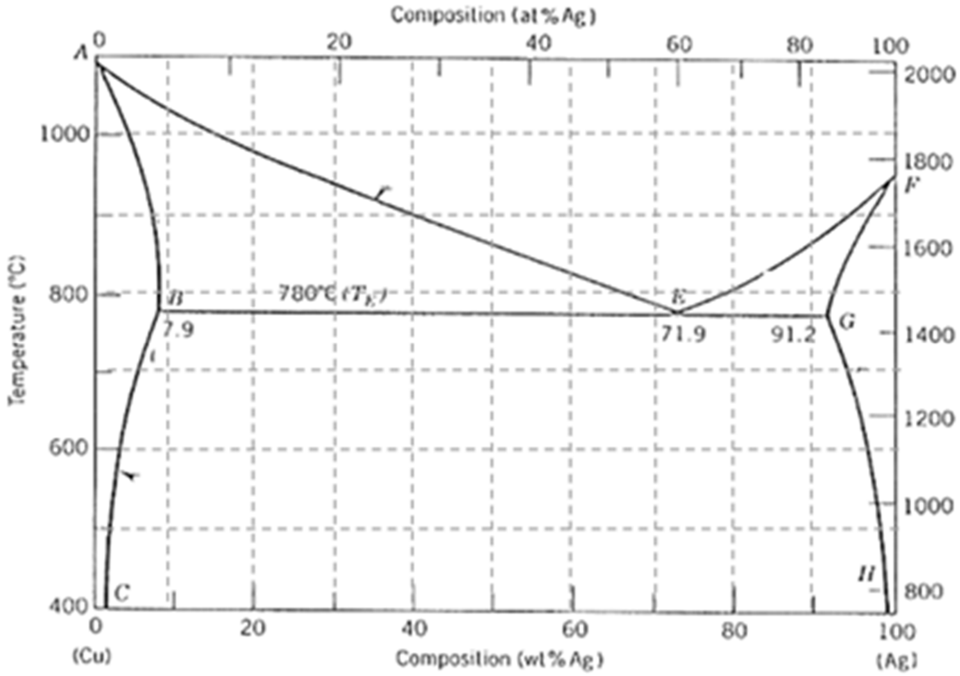
Name and surname……………………………………………… Matricular number……………

Simulation of the Exam "Science and technology of materials"

Important notes: Exam duration: 45 min (9 questions). In multiple choice questions: wrong answer penalty: –0.25 points: answer not given: 0 points. Write the answers to the open questions and the resolutions of the exercises, numbering them, on the attached sheets. Write in an understandable way.

1) Complete the Cu-Ag state diagram shown in the figure indicating the phases in the various regions. Consider an alloy containing 80 wt.% of Ag:
a) indicate which phases are present at 1000 ° C, their composition, the relative amounts and the variance of the system;
b) indicate which phases are present at 600 ° C, their compositions, the relative amounts and the variance of the system.
[5 points]

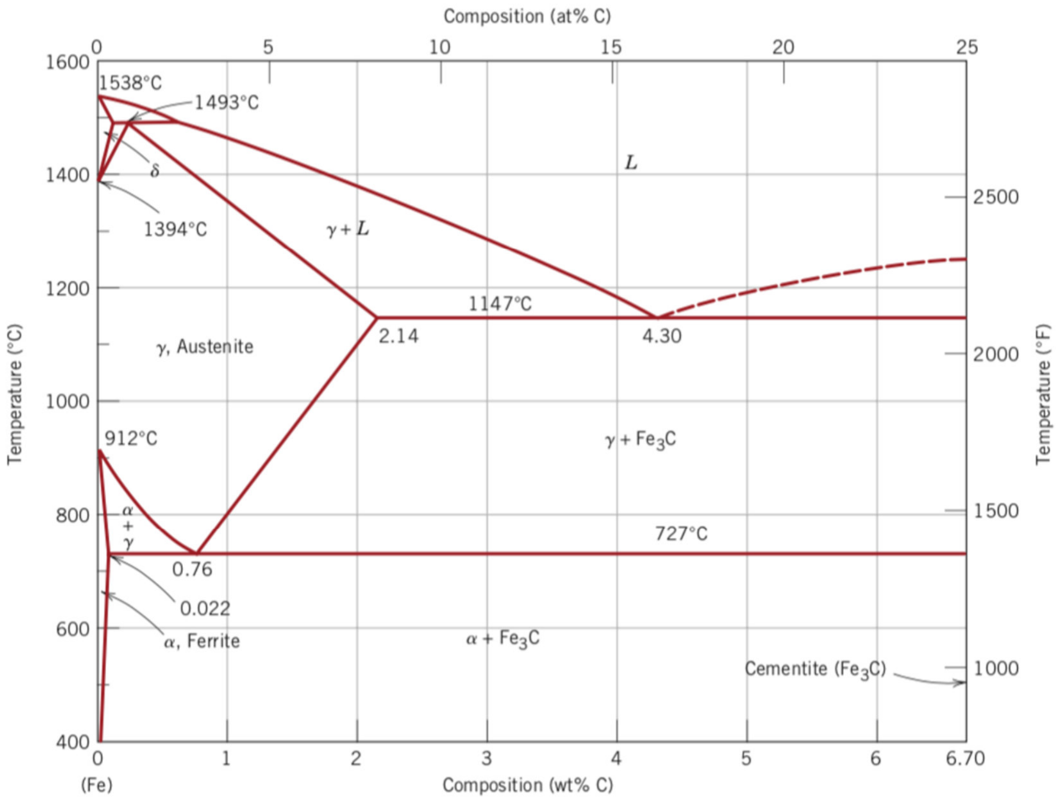


2) Explain the mechanisms of heat conduction in solids. Define the coefficient of thermal conductivity (k), its units and list k of polymers, metals, glasses and crystalline ceramics; also explain the list order. Why diamond is a good thermal conductor?
[points 5]

3) Describe the process of sintering, also explaining to which materials is applied and why.
[points 3]

4) Calculate the elongation (in mm) of a 2 m long steel rod (CTE =14 × 10^{-6} °C^{-1}) when the temperature increases from 20 to 100 °C.
If the same rod is bound at its ends, calculate the thermal stress developed in it, specifying the type (tension, compression, torsion…). The elastic modulus of steel is E = 210 GPa.
[points 3]

5) Calculate the primary (pro-eutectoid) cementite (wt.%) at 400°C for a steel with 1.5% C.
What is pearlite and where can you find it in the phase diagram (if possible)?
What is martensite and where can you it in the phase diagram (if possible)?
[points 5]



6) Porosities in ceramic materials (only 1 correct item):
(a) reduce the tensile strength if they are open and interconnected, while they have no effect on the mechanical properties if the pores are closed
(b) are useful for increasing the compressive strength
(c) are formed during the sintering process
(d) reduce the tensile strength if their characteristic size is less than 1 nm
(e) contribute to increasing thermal insulation
[points 1]

7) If an interstitial atom is added into a crystalline lattice above its solubility limit, we have the formation of (only 1 correct item):
(a) a solid substitution solution
(b) an interstitial solid solution and a substitutional solid solution
(c) an interstitial solid solution and a low-melting liquid phase
(d) an interstitial solid solution and a glassy phase at the grain boundaries
(e) none of the previous responses
[points 1]

8) A specimen made of an unknown material (initial diameter 16 mm, initial length 75 mm) is subjected to a tensile test; the resulting force-elongation curve is shown in the figure below. At the end of the test, the diameter is equal to 11 mm.

You are required to:

a) determine the stress at break;

b) determine the yield strength at 0.2% offset;

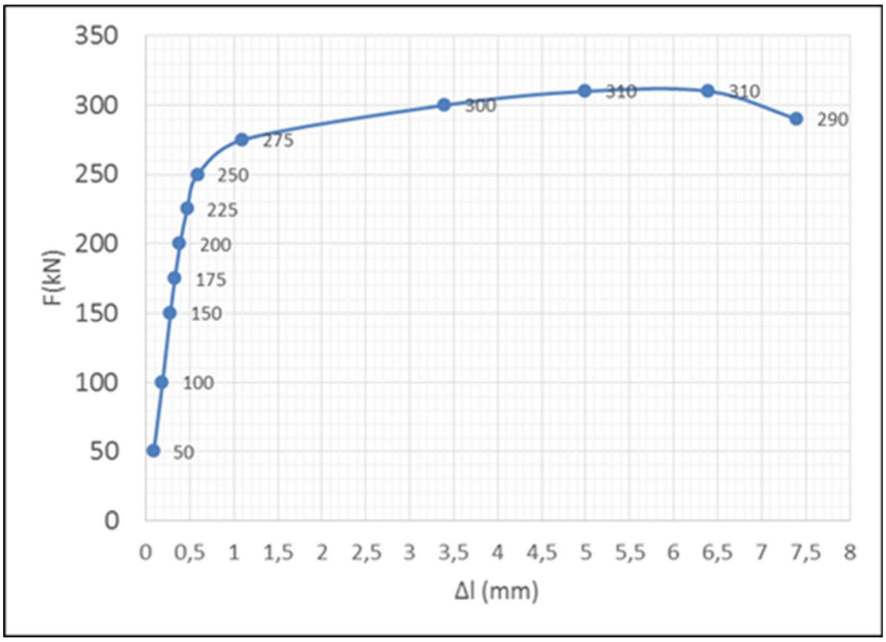
b) determine the elastic modulus;

c) identify the class of material (metal, ceramic, polymer, composite) and hypothesize its type, justifying the response;

d) determine the deformation at break;

e) quantify the necking and describe this phenomenon.

[points 6]



9) Which one of the following tensile stress/strain curves corresponds to a thermoset polymer? What are the main differences between thermoplastic and thermoset polymers?

[points 3]